



सत्यमेव जयते

NITI Aayog

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Sector at a Glance

Automotive Manufacturing

Manufacturing

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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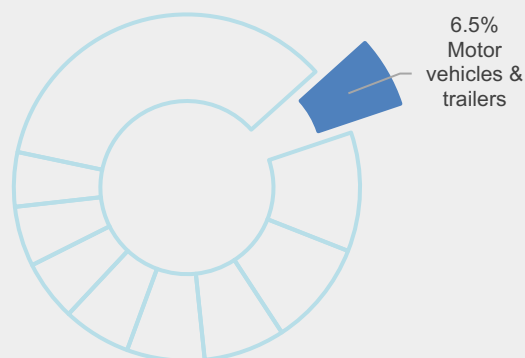
Automobile industry in India



The automobile sector one of the major sectors in the Indian economy, an early success story of manufacturing in India, contributes **9.8%**¹ of India's manufacturing GVA and employs over **19 million people**², which make up **~6.5%**² of overall employment share.

Gross Value Added (India)

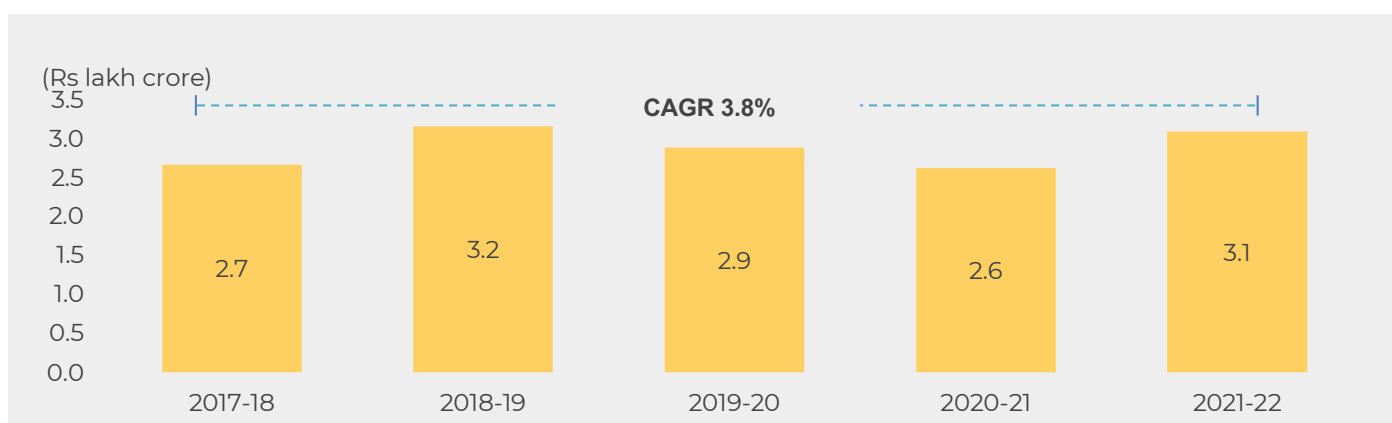
Employment share of automobile sector



Source: Ministry of Statistics and Program Implementation

Source: Annual Survey of Industries 2019-2020

Figure 1: Gross Value Added by automobile and transport equipment sector



Source: Ministry of Statistics and Program Implementation

Gross Value Added (GVA) by the automobile and transport equipment industry is estimated to have grown at a compound annual growth rate (**CAGR**) of **3.8%** over fiscals 2018-2022 with GVA of fiscal 2022 at **3.1 trillion**². This GVA is estimated to have surpassed pre-COVID-19 levels in fiscal 2023.

The automobile industry can be further classified into following subsegments:

1. Passenger vehicles (PV)
2. Commercial vehicles (CVs)
3. Two-wheelers
4. Tractors

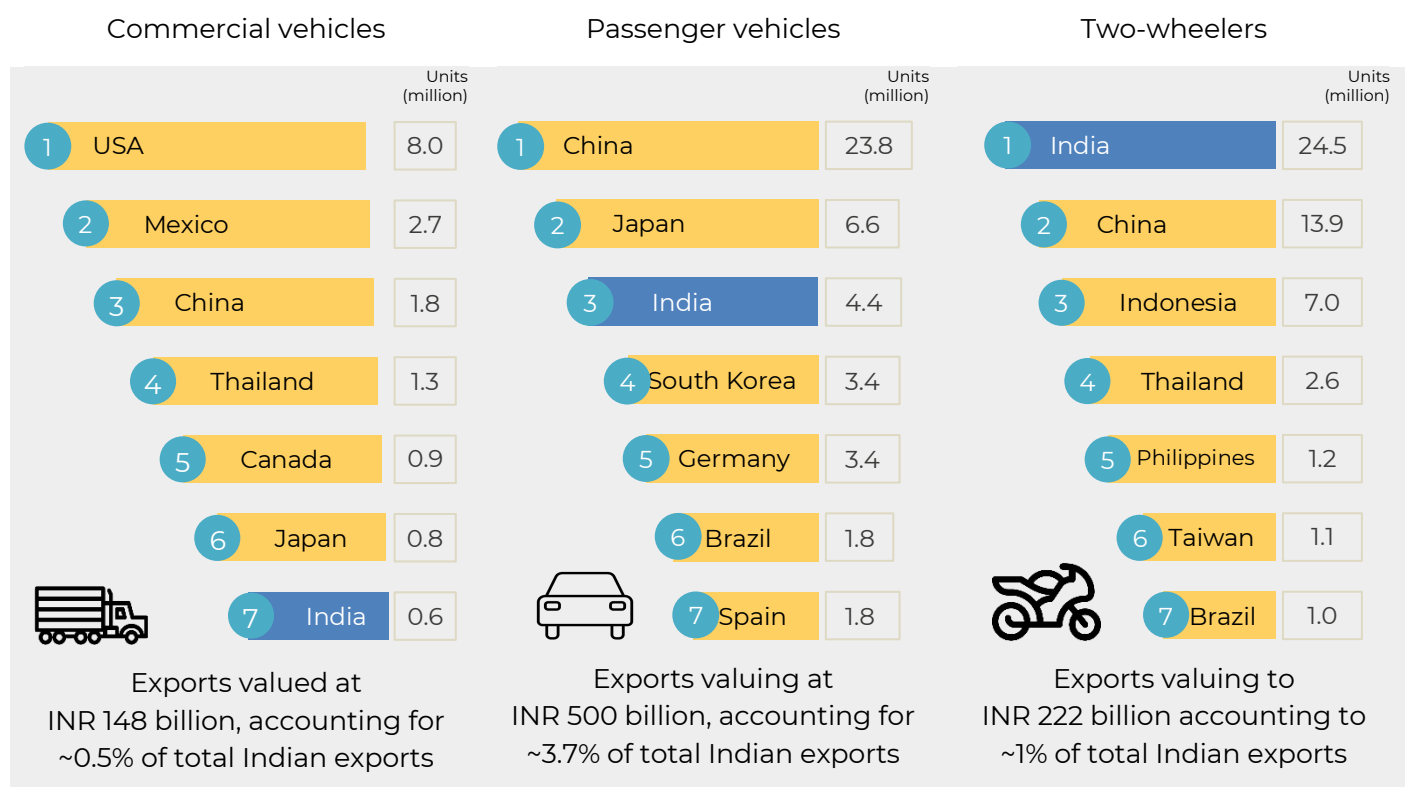
¹ Ministry of Statistics and Program Implementation

² Annual Survey of Industries 2019-2020

Global automobile production benchmarking

In fiscal 2022, India was the 7th largest market for Commercial vehicle production at about 10% of the largest market, the USA. India was the third largest market for Passenger vehicle production in fiscal 2022 after China and Japan which signifies the early success story of automobile manufacturing in India with the Indian Passenger vehicle sector a mix of Indian OEM's, Foreign OEM's and partnerships. India is the largest two-wheeler manufacturer in the world on account of a large population of 1.4 billion people with low per-capita income of Rs 1,72,000 in fiscal 2023 coupled with poor public transport infrastructure and well-established Indian OEM's who gleaned manufacturing chops by partnering with global OEM's.

Figure 2: India tops the charts in two-wheeler production

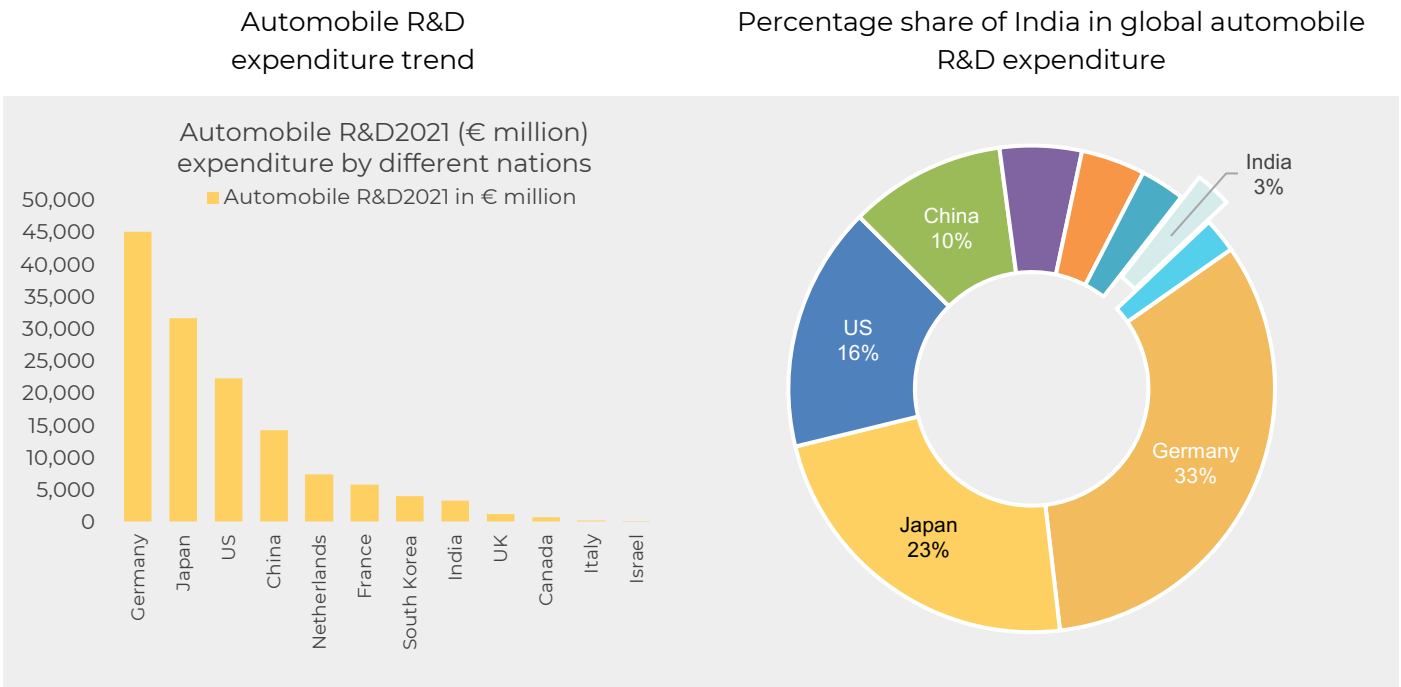


Note: Commercial vehicle and passenger vehicle production numbers are as of FY22; two-wheeler production numbers are as of FY18, OECA classification vehicle category differs with SIAM classification

Source: OECA, Industry

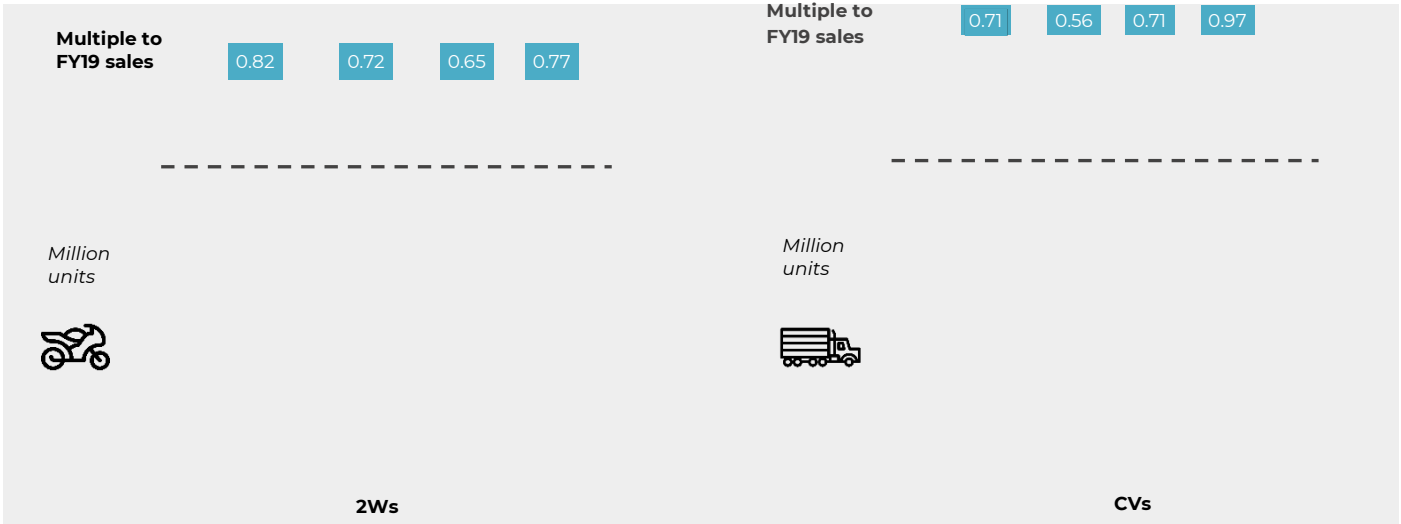
India's allocation of just 2% of the global automotive R&D budget in 2021 indicates significant room for growth and innovation. While established automotive powerhouses like Germany, Japan, the US, and China dominate the R&D landscape with substantial outlays, India's relatively lower share underscores the need for increased investments and focus on research and development in the automotive sector. As India aims to become a global automotive hub, boosting R&D funding and fostering innovation can be a critical step in achieving competitiveness and sustainability in the ever-evolving automotive industry especially with the sector at the cusp of a transition to alternate mobility.

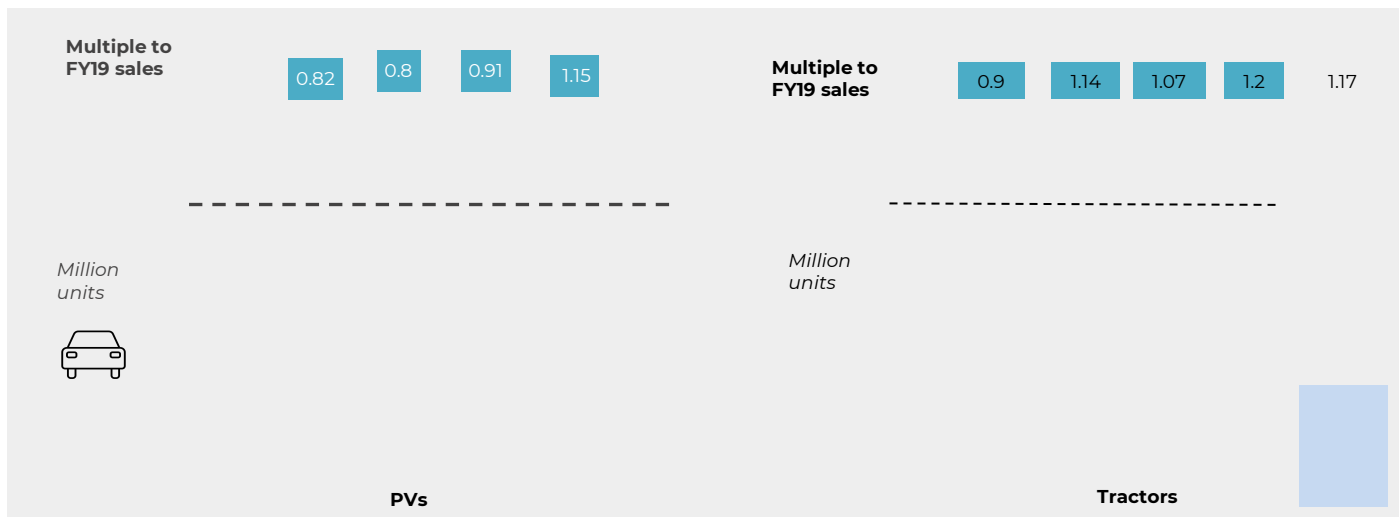
Figure 3: Automobile expenditure on R&D by different nations



Source: IRI (European commission)

Figure 4: Automobile sub-sector sales snapshot





Numbers in boxes indicate volume multiplier as a proportion of fiscal 2019 levels
 Source: Society of Indian Automobile Manufacturers (SIAM), Tractor and Mechanization Association (TMA), CRISIL MI&A Research

Fiscal 2019 was the best year across automobile sectors with sales peaking across segments on account of healthy economic growth. A number of factors post fiscal 2019 led to a drop in sales across sectors. Chief amongst them was the Covid-19 pandemic in fiscal 2021 which led to reduced mobility and impacted incomes, the primary drivers for passenger vehicles, tractors and two wheelers, impacting sales. Other factors were sharp rise in vehicle prices, especially for two wheelers with prices rising by 40% between fiscals 2019 and 2023 on account of regulatory norms - Bharat Stage IV to Bharat Stage-VI transition, compulsory three-year third-party insurance for two wheelers, safety norms – compulsory headlight On, Front airbags, general economic slowdown coupled with a hike in commodity prices. Only in fiscal 2023 did passenger vehicle sales surpass pre-Covid highs of fiscal 2023 aided by improving supplies, low interest rates, a K shaped economic recovery and pent up replacement demand. Two-wheeler sales recovered only to 77% of pre-Covid levels in fiscal 2023 on account of the aforementioned reasons with incomes still to catch up to hike in prices. Commercial vehicle sales recovered to 97% of pre-Covid levels aided by actualization by pent up replacement demand aided by improving freight rates and transporter profitability. Tractor sales improved on-year in fiscal 2021 on account of reverse migration, agriculture activities being exempted from Covid-19 lockdowns and fewer alternate avenues to spend. Sales improved further in fiscal 2023 aided by healthy farm commodity prices pushing up farmer incomes.

In fiscal 2024, all segments expect two wheelers are expected to surpass their pre COVID sales record with Tractors and Passenger vehicles at the vanguard.

Automobile manufacturing hub



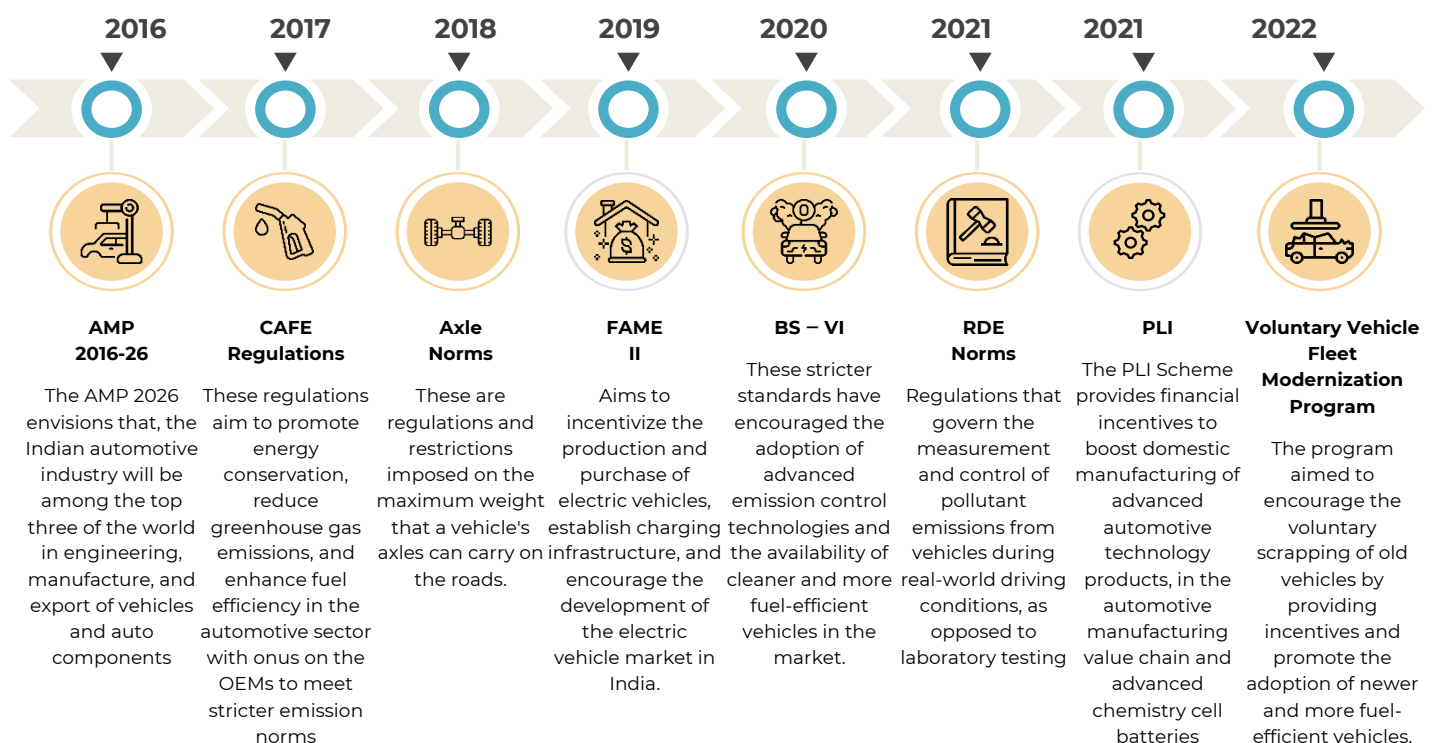
Legend:



Source: Industry, Crisil MI&A research

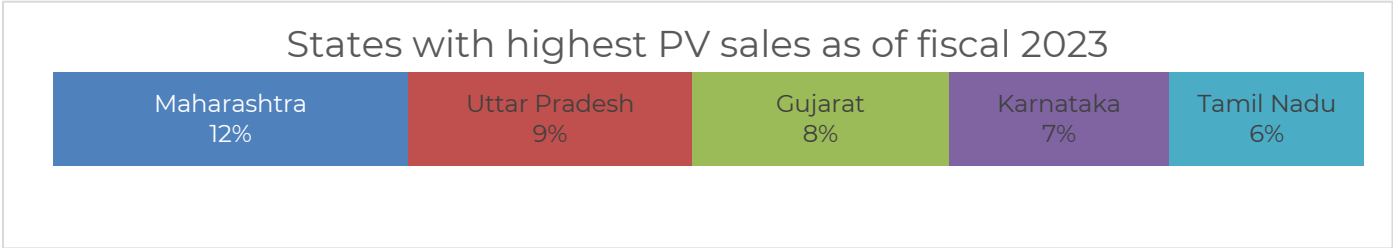
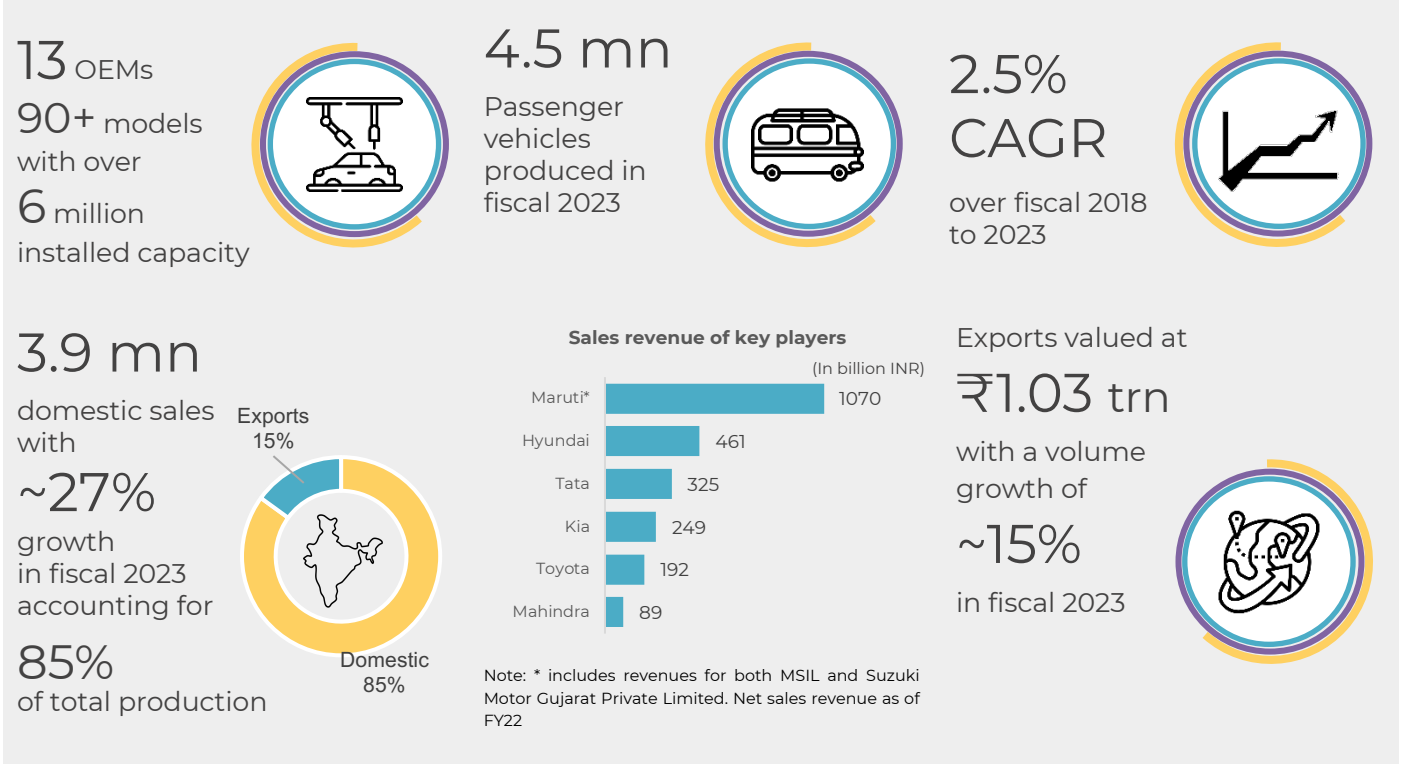
- Tamil Nadu and Maharashtra have the largest production capacity for passenger vehicles.
- Bangalore – Chennai belt has the highest commercial vehicle production in India.
- Two-wheeler production concentrated around Maharashtra, Tamil Nadu, and NCR region.
- New entrants have a slew of capacity additions planned, in addition to the existing OEMs expanding capacities for EVs

Major Policies and Norms



Passenger vehicles overview

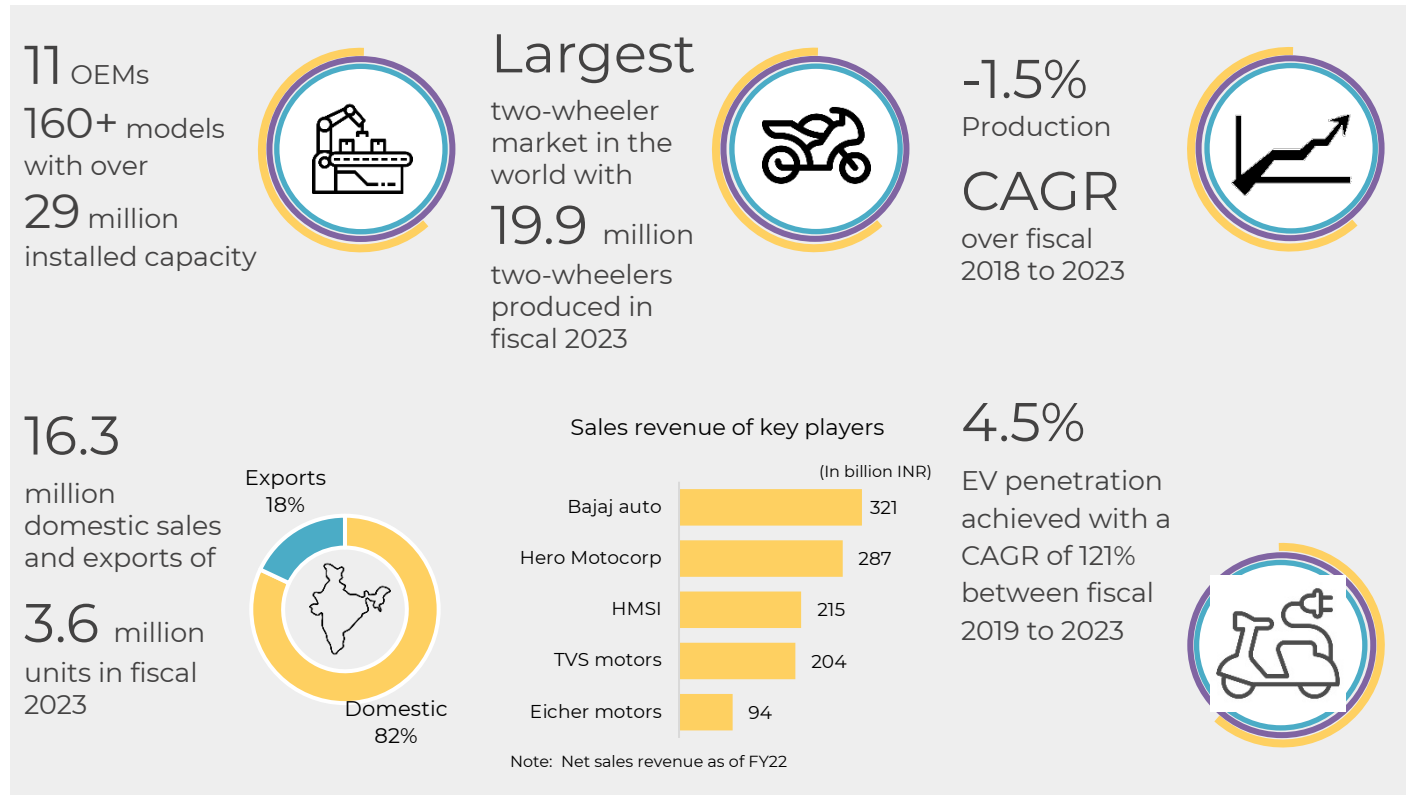
The Indian passenger vehicle industry was the 3rd largest by production in fiscal 2022. It consists of Indian OEM's, foreign OEM's as well as partnerships between the two with 13 OEMs offering over 90 models. The sector boasts an installed capacity of over 6 million vehicles, with an annual production of 4.5 million passenger vehicles in fiscal 2023. The industry has achieved a 2.5% CAGR from fiscal 2018 to 2023, further fueling its prominence in the global automotive landscape. Moreover, India's passenger vehicle exports, valued at ₹1.03 trillion, witnessed a remarkable volume growth of approximately 15% on-year in fiscal 2023, reflecting the industry's international competitiveness and positive outlook.



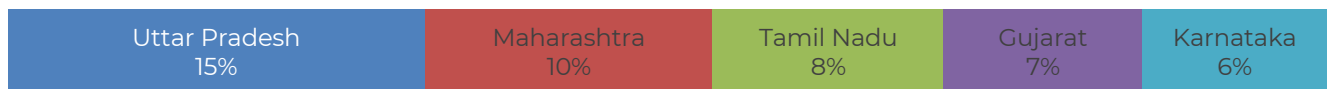
Source: Society of Indian Automobile Manufacturers (SIAM), Ministry of Commerce and Industry

Two-wheeler Overview

The Indian two-wheeler industry is the largest in the world with 11 OEMs offering an extensive lineup of 160+ models and an installed capacity of 29 million units. India retained its position as the world's largest two-wheeler market, producing 19.9 million units in fiscal 2023, alongside domestic sales and exports totaling 16.3 million and 3.6 million units, respectively. Notably, the industry achieved a notable 4.5% electric vehicle (EV) penetration in fiscal 2023 with an impressive CAGR of 121% between fiscals 2019 to 2023, reflecting a significant shift towards sustainable mobility in this high-growth sector.



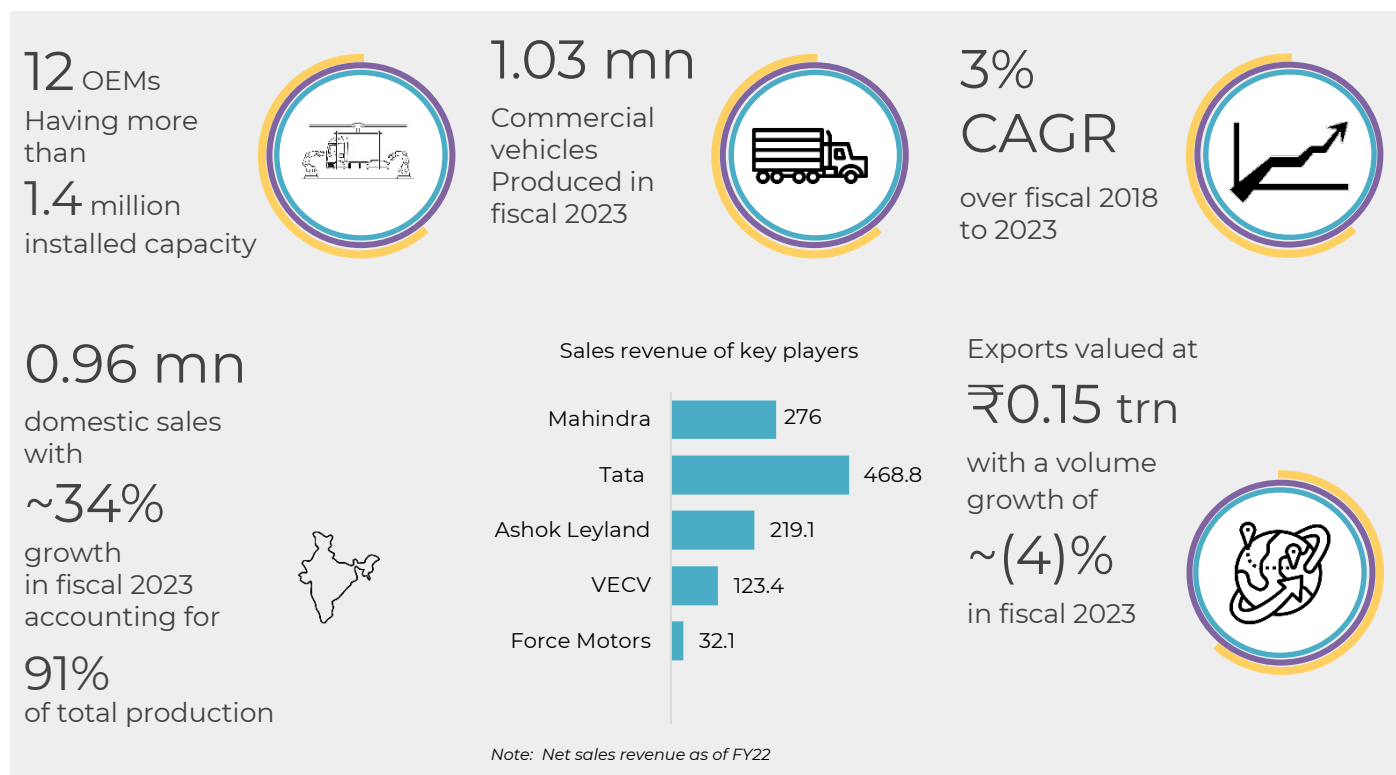
States with highest two wheeler sales as of fiscal 2023



Source: Society of Indian Automobile Manufacturers (SIAM), Ministry of Commerce and Industry

Commercial vehicles overview

The Indian commercial vehicle industry comprises of Light commercial vehicles, medium and heavy commercial vehicles and buses, comprising 12 OEMs with an installed capacity exceeding 1.4 million units. In fiscal 2023, the industry produced 1.03 million commercial vehicles, registering a 3% CAGR from fiscal 2018 to 2023. Notably, domestic sales reached 0.96 million units, exhibiting a ~34% on-year growth, representing a significant portion at 91% of total production. In parallel, exports remained valuable, amounting to ₹0.15 trillion, despite a slight ~4% volume decline in fiscal 2023, underscoring the industry's role in global trade.



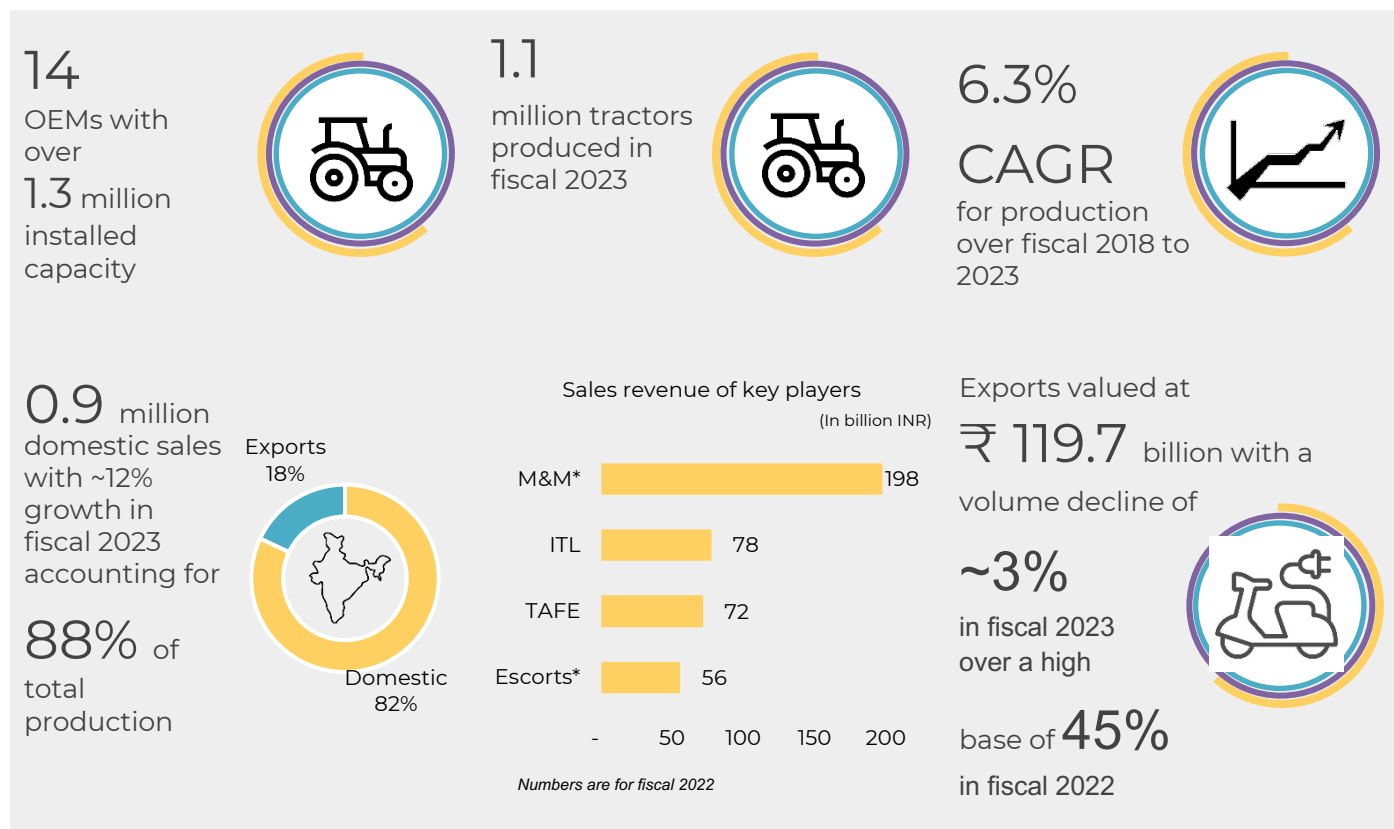
States with highest CV sales as of fiscal 2023



Source: Society of Indian Automobile Manufacturers (SIAM), Ministry of Commerce and Industry

Tractors Overview

the Indian tractor industry encompasses 14 OEMs with an installed capacity of over 1.3 million units. In fiscal 2023, the industry produced 1.1 million tractors, achieving a commendable 6.3% CAGR in production from fiscal 2018 to 2023. Domestic sales reached 0.9 million units, with a ~12% on-year growth in fiscal 2023 reaching all-time high numbers, constituting a substantial 88% of total production. While exports were valued at ₹119.7 billion, they experienced a marginal ~3% volume decline in fiscal 2023 compared to the high base set in fiscal 2022.



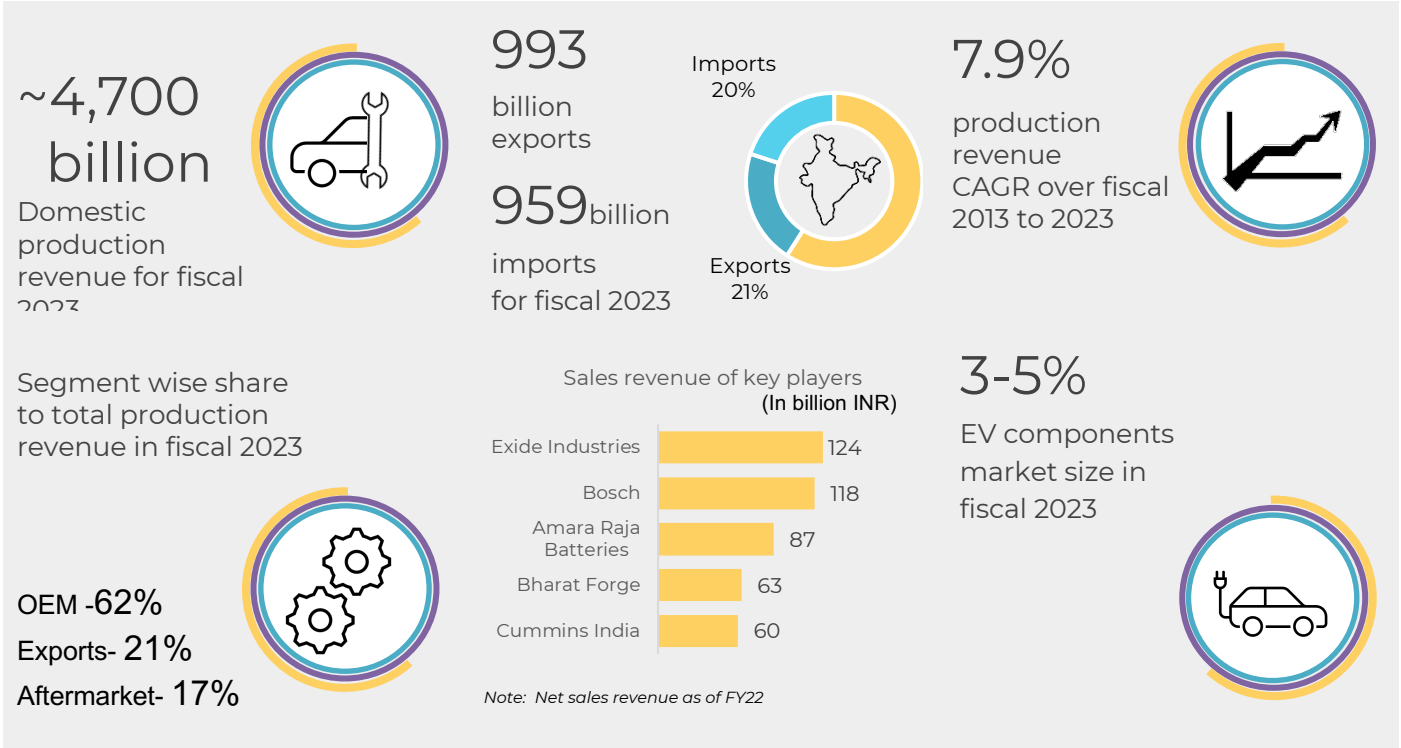
States with highest tractor sales as of fiscal 2023



Source: Tractor and Mechanization Association (TMA), Ministry of Commerce and Industry

Auto components Overview

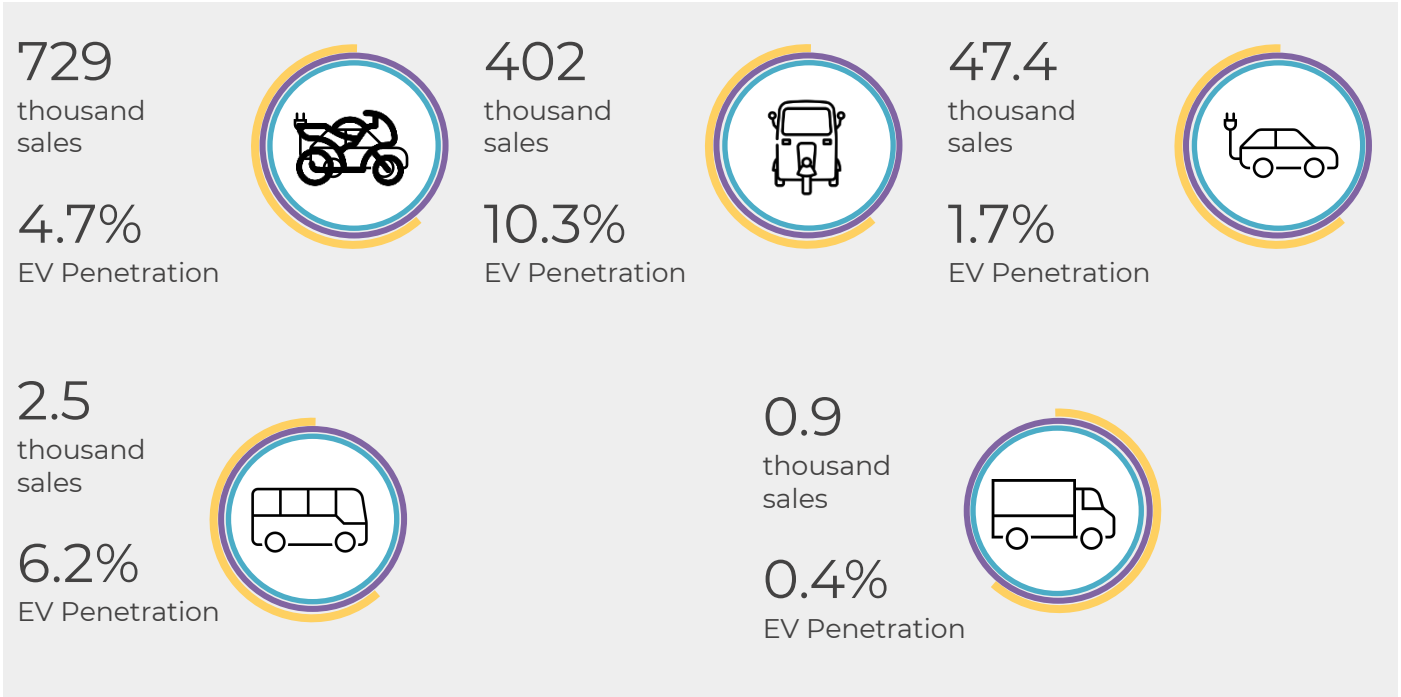
The Indian automotive component industry generated approximately ₹4,700 billion in domestic production revenue with OEMs dominating with a substantial 62% share, followed by exports at 21%, and the aftermarket at 17%. Furthermore, the industry contributed to India's global trade, with exports totaling ₹993 billion, while imports reached ₹959 billion in fiscal 2023. Over the past decade, the sector demonstrated an impressive 7.9% production revenue CAGR from fiscal 2013 to 2023, emphasizing its sustained growth and significance in the country's automotive ecosystem.



Source: CRISIL MI&A Research, Ministry of Commerce and Industry, Company reports

Electric vehicles Overview

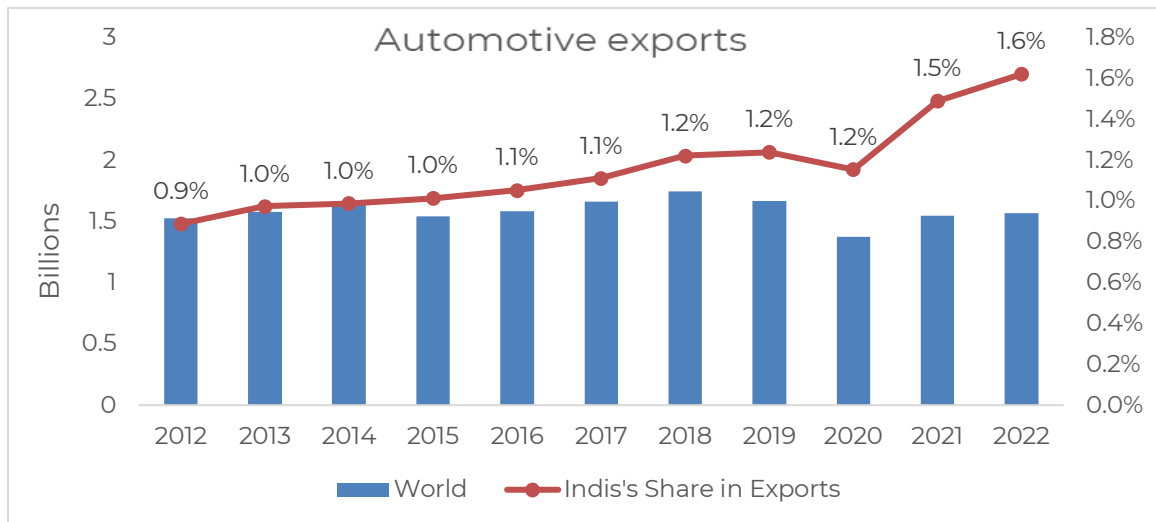
The Electric vehicle (EV) industry is a nascent one in India, with 729,000 sales, reflecting a 4.7% penetration in the two-wheeler market in fiscal 2023. In the three-wheeler segment, 402,000 EVs were sold, achieving a 10.3% penetration rate in the same period. Additionally, the passenger vehicle market saw 47,400 EV sales, accounting for 1.7% of the market share in fiscal 2023. The bus segment, though smaller, exhibited notable growth with 2,500 sales and a 6.2% EV penetration rate, highlighting the promising future of electric mobility in India. Electric vehicle sales are aided by incentives by both the central, under the FAME-2 scheme as well as individual state governments under their respective state government policies.



Note: Data for fiscal 2023
Source: Industry, VAHAN Portal

Opportunities

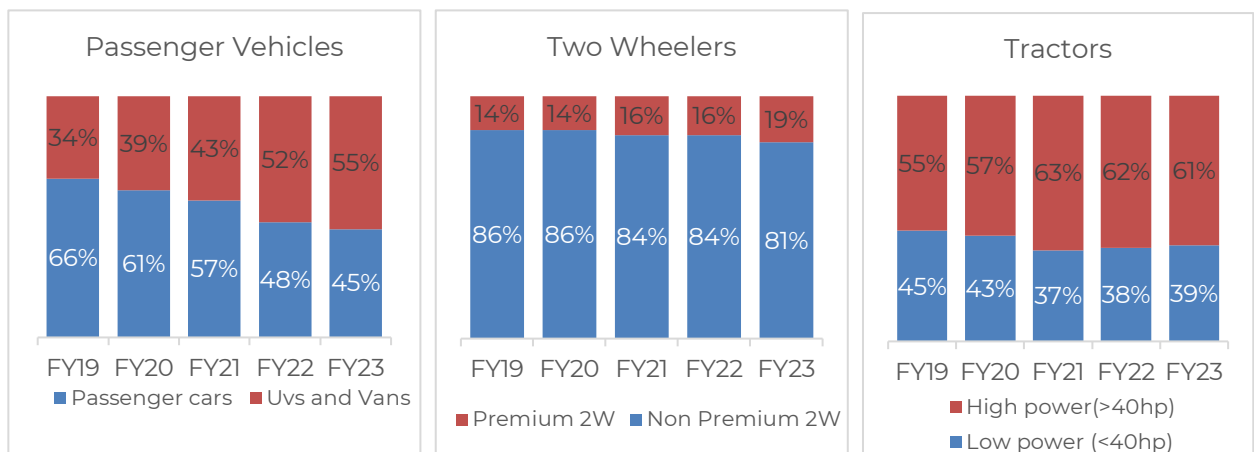
1. Exports growth potential



Source: International Trade Center

Despite Indian automobile exports growing at a CAGR of 6.5% during calendar years 2012 to 2022, against a meagre 0.3% CAGR of overall automotive exports, (consisting of PV, CV, 2W and automotive components), the Indian automotive exports only account for ~1.6% of global automotive exports. This gives Indian automobile players a growth headway with the ongoing transition to electric coupled with the supply chain diversification at play and India with a well-established automobile OEM and auto-components industry seems well set to benefit.

2. Drive to Premiumisation:



Source: Society of Indian Automobile Manufacturers (SIAM), Tractor Manufacturers Association Crisil MI&A research

Rising share of SUVs (55% of overall domestic sales as of fiscal 2023) against once dominant hatchback segment in passenger vehicle segments (45% of overall domestic sales as of fiscal 2023) along with increasing share of high-priced two wheelers and high-powered tractors signifies shifting Indian consumer sentiments towards more premium options. This gives OEMs an opportunity to introduce models towards the higher end of price spectrum with cutting edge safety, technology, and democratization of features. This shift has been enabled by rising average incomes and heightened aspiration of average Indian buyers. This aids in

making Indian vehicles competitive with global peers supporting export potential, improves realisations for OEM's, improves manufacturing chops of OEM's and provides a boost to Indian automotive component players.

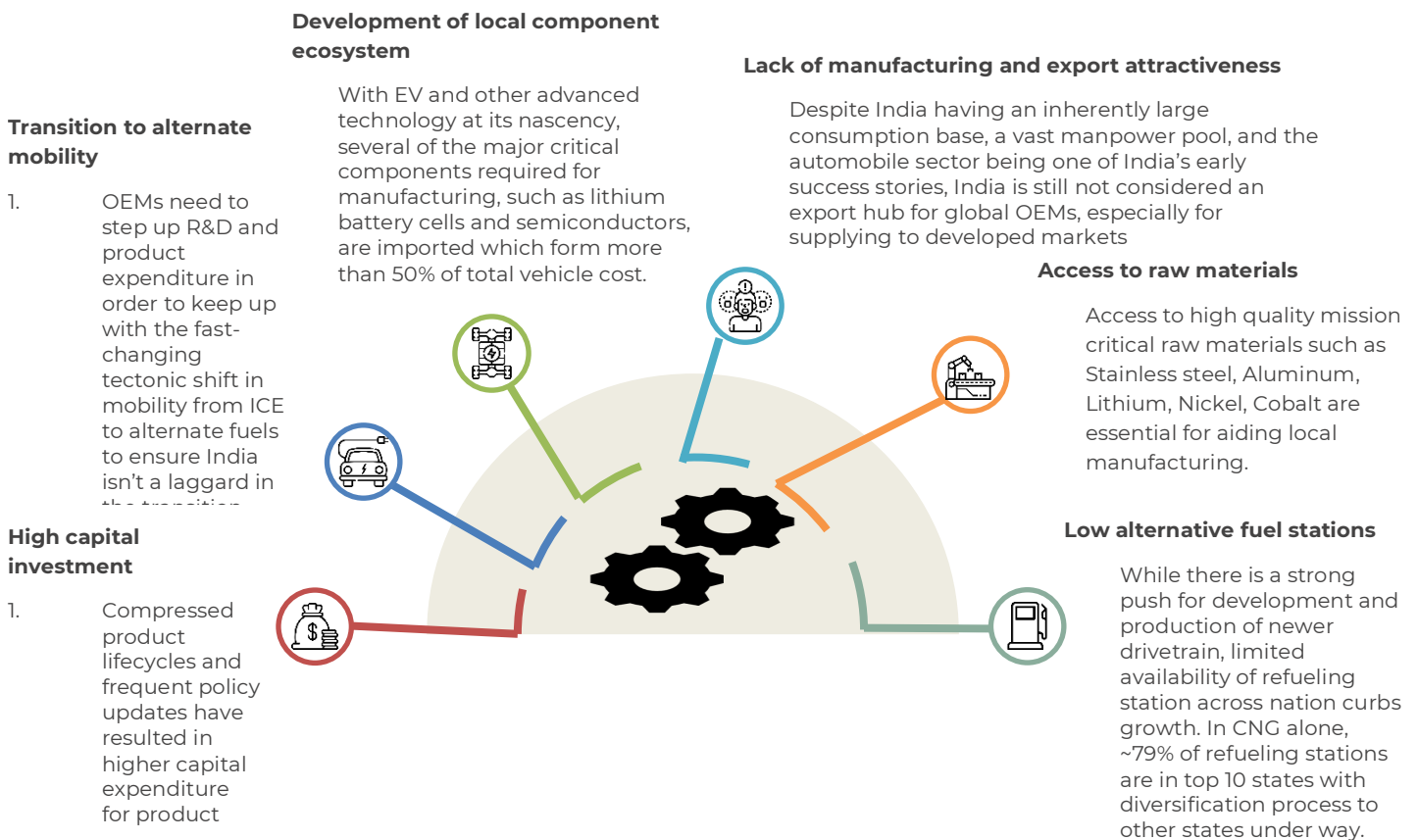
3. EV Manufacturing leadership

As climate risk takes center stage and both advanced and emerging nations pivot toward electric vehicles (EVs), India finds itself in a distinctive position to become a leader in EV production. Innovative government policies like the Faster Adoption and Manufacturing of Hybrid and Electric Vehicles (FAME) initiative have been instrumental in boosting the domestic EV market. Additionally, the Production Linked Incentives offer crucial support to automotive component players, fostering the local manufacturing of EV components. These strategic measures position India to capitalize on the growing global demand for EVs and strengthen its foothold in the rapidly evolving electric vehicle industry.

4. Supply chain diversification post the Covid-19 pandemic

As nations worldwide decrease reliance on Chinese goods, especially in the automotive sector with measures like anti-dumping duties and import bans, India stands poised to leverage its expanding automotive manufacturing sector. Boasting cost-competitive labour, a skilled workforce, and a thriving technology industry, India emerges as an appealing alternative supplier. With a vast domestic market and initiatives such as "Make in India" and the "Production-Linked Incentive (PLI) Scheme," India actively promotes domestic manufacturing and investments. These factors strategically position India as a key partner in the global movement steering away from Chinese goods, facilitating the diversification of the global supply chain.

Challenges



1. High capital investment

The capital-intensive nature of India's automotive industry is exacerbated by the shrinking average holding period of passenger vehicles, dropping from 15 years in the early 2000s to six years as of fiscal 2023. This trend challenges automakers to maintain customer loyalty and maintain profitability. While that entails investment by OEM's for model refreshes and new model launches, it also enables improved offtake for vehicles as customers change vehicles quicker. Furthermore, frequent policy updates enforcing stringent emission and safety standards keep OEM's on their toes. For instance, the transition to Bharat Stage VI emission norms has prompted significant investments in technology upgrades and production processes across the industry. This regulatory change intensifies financial pressures on the industry, requiring a delicate balance between compliance efforts and maintaining competitiveness in a challenging market. With the ongoing transition to electric mobility, OEM's have to walk a tightrope between updating/introducing current ICE models and also undertaking development for Battery Electric Vehicles (BEV) while keeping abreast of other potential shifts to alternate mobility such as Hydrogen. Leading OEMs, including TATA Motors, Maruti Suzuki, and Mahindra & Mahindra, have disclosed substantial capital expenditures for electric vehicle (EV) manufacturing in India. TATA Motors committed INR 6,000 crores, Maruti Suzuki earmarked INR 1,25,000 crores, and Mahindra & Mahindra allocated INR 18,000 crores. In the two-wheeler sector, Hero MotoCorp is investing INR 1,500 crores to enhance its electric vehicle portfolio.

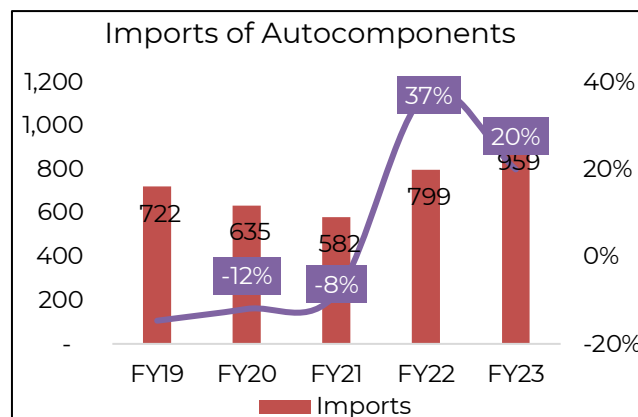
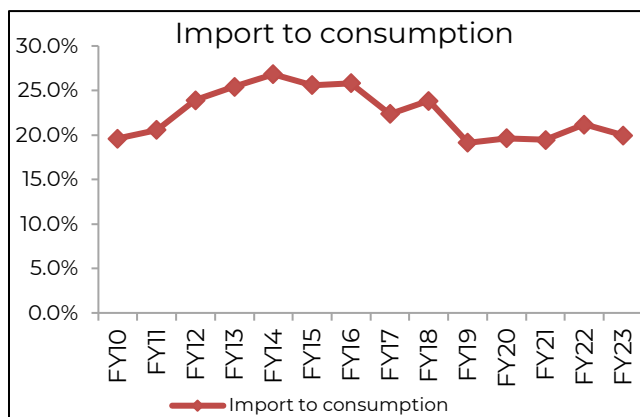
2. Low R&D in transition to alternate mobility

In 2021, India accounted for 2% of global automotive Research and Development (R&D), compared to 8% for the United States, 5% for Germany, and 4% for China, Japan, and South Korea each. This indicates varying degrees of global participation. Given the ongoing shift away from Internal

Combustion Engine (ICE) vehicles, reliance on foreign technologies due to limited R&D investment presents challenges for achieving self-sufficiency in the automotive sector. Bridging this R&D gap is crucial for India to actively contribute to and lead advancements in the dynamic global automotive landscape.

3. Development of local component ecosystem

The absence of a robust local ecosystem for electric vehicle (EV) component manufacturing poses a significant challenge to the industry's growth in India. While the country is making strides in EV adoption, the lack of a comprehensive ecosystem means that crucial components such as lithium-ion batteries, electric motors, and semiconductors are predominantly imported. This reliance on foreign suppliers not only escalates costs but also exposes the industry to supply chain vulnerabilities and geopolitical uncertainties. Developing a strong domestic ecosystem for EV component manufacturing is essential for fostering innovation, reducing dependency on external sources, and ensuring the long-term sustainability and competitiveness of the Indian electric vehicle market.



Source: Automotive Component Manufacturers Association of India ,Crisil MI&A research

The reduction in import share, initially driven by localization efforts supported by government incentives, has reversed due to increased electric vehicle (EV) production. India's reliance on imports for crucial EV components, such as batteries and motors, has surged.

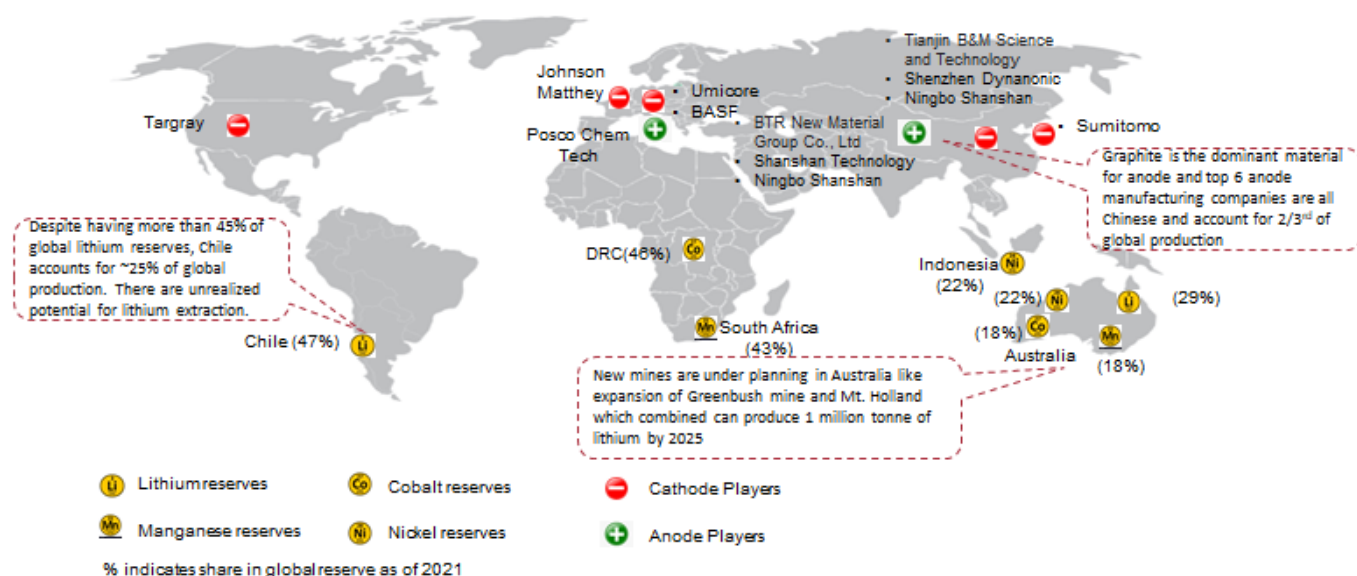
4. Access to raw materials

India's electric vehicle (EV) industry faces a supply challenge as it heavily relies on imports for crucial raw materials essential in EV component manufacturing. Notably, minerals like lithium and rare earth metals, integral to battery production and other key components, are predominantly sourced from abroad. This heavy reliance on imports exposes the industry to global supply dynamics, influencing factors such as geopolitical tensions, trade policies, and fluctuations in commodity prices. For instance, the concentration of critical minerals in a few countries, like lithium in South America and rare earth metals in China, raises concerns about supply chain vulnerabilities. Any disruptions in the supply chain, be it due to geopolitical events or market dynamics, can significantly impact the cost and availability of these essential materials, thereby affecting the growth and sustainability of India's EV manufacturing sector. As the demand for EVs continues to rise, there is an increasing urgency for strategic measures to address these supply challenges and promote domestic production of critical raw materials.

Mineral reserves concentrated majorly in Australia, Africa and South American countries

Anode and cathode manufacturers are spread in Europe and China but China accounts for more than 60% of global supply

Congo (DRC) supplies over 70% of global Cobalt supply while Australia supplied over 50% of global lithium supply



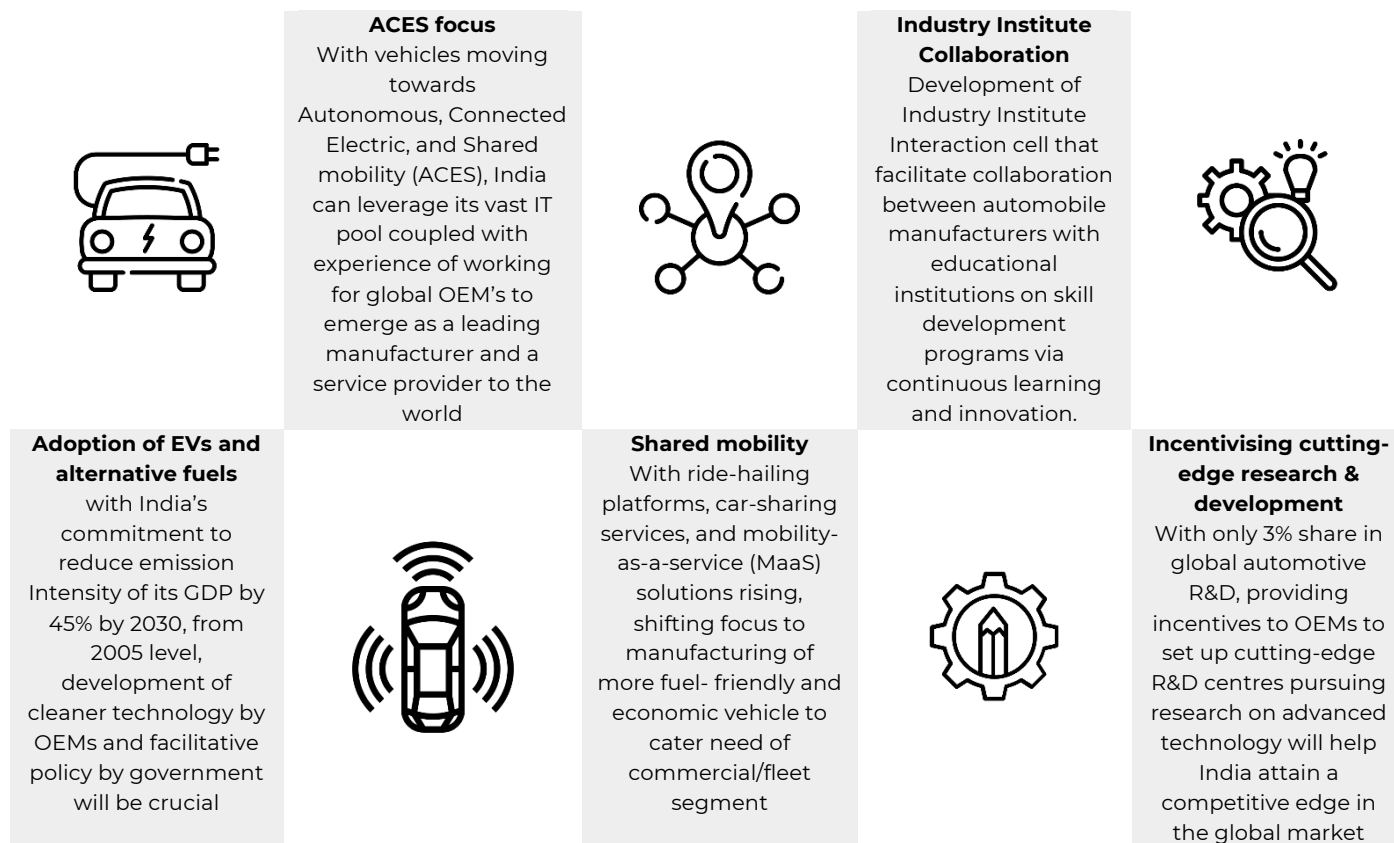
5. Lack of manufacturing and export attractiveness

Despite the advantages of low power and labor costs, India has struggled to attract significant investments, constituting only 1% of the global automotive capital expenditure (Capex) as of 2021. Contributing factors include high logistic costs, accounting for 14% of GDP in India compared to 8-9% in advanced economies. Furthermore, low productivity, with hourly wages at \$9 compared to \$71 in the United States, \$14 in China, and \$11 in Vietnam, has diminished India's appeal for manufacturing and exports. Insufficient spending on research and development (R&D) compounds these challenges, collectively impacting India's competitiveness and global attractiveness in the manufacturing sector.

6. Limited Availability of Alternative Fuel Stations in India

The pursuit of developing and manufacturing advanced drivetrain technologies in India confronts a substantial hurdle due to the scarcity of refueling stations nationwide. This challenge persists even in established fuel types like compressed natural gas (CNG), where approximately 79% of refueling stations are concentrated in the top 10 states. Looking into the future, similar challenges are anticipated for the expansion of hydrogen and liquefied natural gas (LNG) networks, as well as the integration of biofuels, underscoring the urgent need for a comprehensive and evenly distributed refueling infrastructure. Furthermore, the growth of electric vehicles (EVs) is impeded by the early stage of EV infrastructure development, marked by high capital expenditure requirements and a lack of standardization in charging protocols. It is crucial to address these infrastructure challenges to facilitate the seamless integration and widespread adoption of diverse and sustainable drivetrain technologies throughout the country.

Way forward



Source: Ministry of Environment, Forest and Climate Change, EU Industrial R&D Investment Scoreboard

1. Adoption of EVs and alternative fuels

With strong commitment to curb emission, India has the ability to transition to cleaner modes of fuel technology the potential to emerge as a global leader especially in the electric two-wheeler industry. With a strong manufacturing foundation and over a dozen e-two-wheeler manufacturers, the country is positioned for growth. Electric two-wheelers have achieved a 4.5% penetration as of fiscal 2023, with sales reaching 727,000 units³¹. India's strength lies in producing a diverse range of e-two-wheelers, from affordable models for emerging markets in Asia to feature-rich options for advanced economies like Europe. Leveraging the scale of the Indian market, both domestic and foreign OEMs are actively designing and producing electric two-wheelers in the country. With strategic investments in infrastructure, technology, and policies, India can capitalize on its versatility to lead the global electric two-wheeler industry, promoting sustainability and innovation.

2. Development of EV supply chain

To facilitate smooth transition to alternate fuel technology and support stable market dynamics, it is essential for India to develop a secure EV supply chain. Establishing a resilient domestic supply chain necessitates international collaborations, domestic research promotion, and incentives for mining

³¹ VAHAN

exploration projects. The acceleration of the extraction process requires streamlined regulatory procedures, fast-tracked approvals with environmental standards intact, and essential infrastructural support such as road access and electricity provision to mining sites. Simultaneously, replicating successful schemes like the Production-Linked Incentive (PLI) for the EV component sector can fuel growth, innovation, and global competitiveness. By creating a conducive environment for domestic EV component manufacturing, India can solidify its global market position, contributing to economic growth and sustainability in the era of electric vehicles. More global initiatives like Minerals Security Partnership (MSP) will help India secure its external supply of critical minerals.

3. Shift to ACES

As the automotive industry progresses towards Autonomous, Connected, Electric, and Shared mobility (ACES), India possesses a significant opportunity to capitalize on its extensive IT talent pool. With a proven track record of working with global Original Equipment Manufacturers (OEMs), the country can position itself as a prominent manufacturer and service provider on the global stage. Leveraging expertise in software development, data analytics, and artificial intelligence, India can play a pivotal role in advancing technologies essential for ACES. Collaborations between IT firms and automotive manufacturers can drive innovation, allowing India to not only meet domestic demands but also export cutting-edge solutions and contribute substantially to the transformation of the global automotive landscape. Through strategic investments, skill development initiatives, and fostering industry partnerships, India can pave the way for a dynamic and influential role in the rapidly evolving ACES ecosystem.

4. Industry Institute Collaboration

The establishment of an Industry Institute Interaction (III) cell is a proactive step towards fostering collaboration between automobile manufacturers and educational institutions. This initiative aims to facilitate skill development programs by promoting continuous learning and innovation. Through this platform, manufacturers can engage with academic institutions to tailor curricula, ensuring that students are equipped with industry-relevant skills. For example, partnerships between automotive giants like Tata Motors or Mahindra and educational institutions in India can lead to the creation of specialized courses on electric vehicle technology or advanced manufacturing techniques. This collaborative approach not only addresses the industry's evolving needs but also provides students with real-world exposure and practical insights, enhancing their employability in the automotive sector. Additionally, the III cell acts as a conduit for joint research projects, internships, and workshops, fostering a symbiotic relationship between academia and industry for mutual growth and advancement.

5. Incentivizing cutting-edge research & development

India's current 2% share in global automotive Research and Development (R&D) underscores the need for strategic measures to enhance its position in the global market. Offering incentives to Original Equipment Manufacturers (OEMs) to establish cutting-edge R&D centers can be a transformative step. By encouraging research on advanced technologies, such as electric vehicles, autonomous driving, and sustainable mobility solutions, India can elevate its automotive industry to a competitive level globally. This approach not only attracts foreign investment but also cultivates a culture of innovation and technological excellence within the domestic automotive sector. For instance, providing tax incentives, research grants, or regulatory support can incentivize OEMs to allocate resources for groundbreaking R&D projects. This, in turn, positions India as a hub for pioneering automotive technologies, fostering economic growth, job creation, and establishing the country as a key player in the rapidly evolving global automotive landscape.

